

Mohammed Khaleel Pasha

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 Hyderabad, India | Willing to relocate

Professional Summary

System Design Engineer (CBTC Signaling) with over 10 years of international experience across wayside-centric moving block and train-centric full moving block architectures. Lead gap analysis between baseline products and project needs, translating user needs into use cases, operational scenarios (ConOps), functional/architectural solution concepts, and controlled specification updates via Change Requests.

CORE COMPETENCIES

- Systems Engineering & Requirements Management.
- Gap Analysis & Solution Concept Development (baseline vs project needs)
- Concept of Operations (ConOps) & Operational Scenario Engineering (nominal/degraded modes)
- System Functional & Architectural Design (interfaces, allocations, constraints)
- Change Request (CR) Impact Analysis & Specification Updates
- Stakeholder Management & Design Reviews (workshops, trade-offs, GO/NO-GO readiness)
- Documentation Baseline Management (SyRS, SyAD, scenarios, technical reports)
- Expert in Headway Simulation, & ATO Profile Optimization.
- Technical Tools: ATS, SCADA, IBM DOORS, MATLAB/Simulink, & AutoCAD.

PROFESSIONAL EXPERIENCE

Alstom Transport India Ltd. | Hyderabad, India

System Designer | Apr 2023 – Present

- Conduct gap analysis between baseline train-centric CBTC product behaviour and project operational needs; translate customer needs into user needs, use cases, and operational contexts.
- Develop operational scenario specifications (nominal + degraded modes) and ensure consistency with functional and architectural viewpoints.
- Lead solution definition workshops: evaluate alternatives, document trade-offs, and present functional + architectural + operational analysis for design GO/NO-GO decisions.
- Prepare and manage Change Requests: impact assessment on scenarios/specs, traceability updates, and baseline documentation alignment
- Authored high-level system engineering deliverables (SyRS/SyAD/ConOps equivalents), documenting functional behaviour, architecture, and operational control concepts for wayside-centric moving block CBTC solutions.
- Conducted advanced headway simulations and ATO profile tuning to ensure punctual and reliable train movements.

Keolis - Hyderabad Metro Rail | Hyderabad, India

Group Station Controller | Oct 2022 – Apr 2023

- Supervised station operations across multiple locations, ensuring compliance with all safety rules and operational procedures.
- Monitored network KPIs and ensured 24/7 operational readiness and continuity.
- Led emergency response planning, coordinating with the Operations Control Center (OCC) to mitigate disruptions.

Station Controller / Local Incident Officer | Oct 2018 – Oct 2022

- Managed traffic and safety for high-traffic interchange hubs (150,000+ commuters/day), utilizing ATS and SCADA systems.

- Acted as Local Incident Officer, coordinating directly with the OCC during degraded circumstances to ensure safe train movements.
- Maintained computerized records of all activities and reported environmental conditions affecting rail operations to central control.

KG Mech Pvt Ltd | Hyderabad, India

MEP Engineer | Apr 2016 – Oct 2018

- Executed MEP engineering tasks for metro infrastructure, coordinating HVAC, plumbing, and electrical system designs.

EDUCATION & ACADEMIC QUALIFICATIONS**Bachelor of Technology (B.Tech) in Mechanical Engineering | 2012 – 2016**

Graduated with Distinction (70.32%). Provides the highly desirable engineering discipline foundation required for deeply comprehending rolling stock mechanics, heavy rail traction systems, and complex rail infrastructure.

PROFESSIONAL QUALIFICATIONS & CERTIFICATIONS**Certified Trainer – KEOLIS:**

Recognized train-the-trainer qualification, validating expertise in instructional delivery, adult learning methodologies, and competency assessment within a rail operations context.

Certified Systems Engineering & Requirements Writing Professional:

UNSW Sydney (The University of New South Wales) – Validates advanced capabilities in designing structured learning materials, e-learning content, and precise competency specifications.

Digital & E-Learning Capabilities:

Highly proficient with modern MS Office tools, AUTOCAD, IBM DOORS, and simulation software (MATLAB/Simulink).

Languages:

Fluent English , Conversational Arabic, Hindi, Telugu, Urdu.